Interdisciplinary (Project Work) Innovations Design and Thinking Lab
IIDTL-1BPRJ258

RFID Staff Tracking System

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Presentation Outline

- Introduction
- Literature Survey
- Problem Statement
- Prototype Project Design
- Methodology and Algorithms
- Software /Hardware Descriptions
- Experimental Results
- Merits/Applications
- Conclusion
- Project Budget
- References

Introduction

Objectives:

- To develop an RFID-based Smart Faculty Latest Entry Updating System.
- To reduce the time spent by students in searching for faculty members.
- To record the latest entry location of faculty using RFID technology.

Scope of the Project:

- Applicable in colleges, universities, and educational institutions.
- Provides information about the faculty's latest recorded location within the campus.
- Enables students to access faculty last-seen information through a web dashboard.
- Improves communication and academic efficiency by reducing search time.

Literature Survey

Sl.No	Title	Authors & Publication Date	Summary
01.	IoT Enabled Smart Campus Management System	M. Patel, K. Shah, P. Desai (2024)	This study proposes an IoT-based campus management framework for monitoring resources and personnel in real time using cloud technology.
02.	Real-Time Personnel Tracking Using RFID and Cloud Computing	J. Lee, H. Kim, S. Park (2022)	The paper discusses RFID-enabled personnel tracking integrated with cloud databases to provide location updates and monitoring capabilities.
03.	Smart Faculty Tracking System Using IoT Technologies	S. Verma, A. Singh, N. Mishra (2024)	The research focuses on faculty tracking in educational institutions using RFID and wireless communication to improve accessibility and campus management.
04.	ESP32 Based RFID Tracking and Monitoring System	P. Reddy, V. Kumar, R. Naik (2023)	The paper demonstrates the use of ESP32 and RFID technology for real-time tracking applications with cloud database integration.
05.	Cloud-Based Location Tracking System for Educational Institutions	T. Johnson, M. Williams (2023)	This work presents a cloud-connected location tracking platform that enables students and administrators to identify the latest location of faculty members efficiently.

Problem Statement

STUDENT



- Needs to meet faculty
- Seeks academic guidance
- Requires project/ lab assistance
- Wants to know the faculty's **latest location and last seen time**

LARGE CAMPUS



- Multiple departments and blocks
- Numerous classrooms, labs and offices
- Faculty members move frequently between locations
- Manually searching takes time

LATEST ENTRY CHALLENGE



- Not a real-time tracking system
- Location is updated only when faculty scans RFID card at a location
- Students may not know where faculty last entered
- Information may be outdated if not scanned recently

OUR SOLUTION (LATEST ENTRY UPDATE SYSTEM)



- RFID cards assigned to faculty
- Scanned at installed RFID readers
- ESP32 captures UID, location and timestamp
- Data uploaded to Firebase Realtime Database
- Shows faculty's latest entry location and last seen time

BENEFITS FOR STUDENTS



- View faculty's **latest recorded location**
- See last seen timestamp
- Saves time and effort
- Improves academic communication
- Helps in planning meetings efficiently

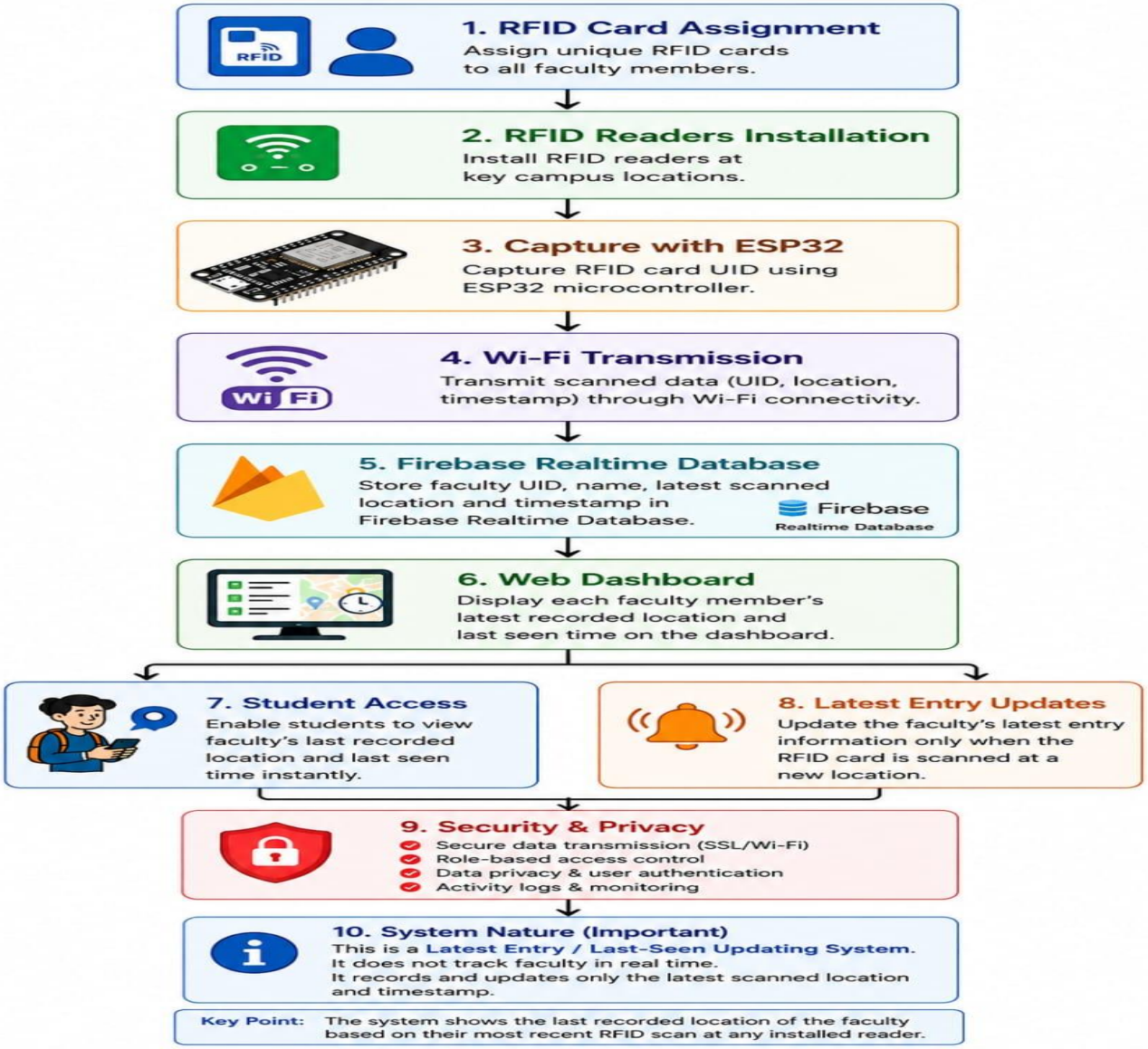
TIME SAVED & EFFICIENCY GAINED



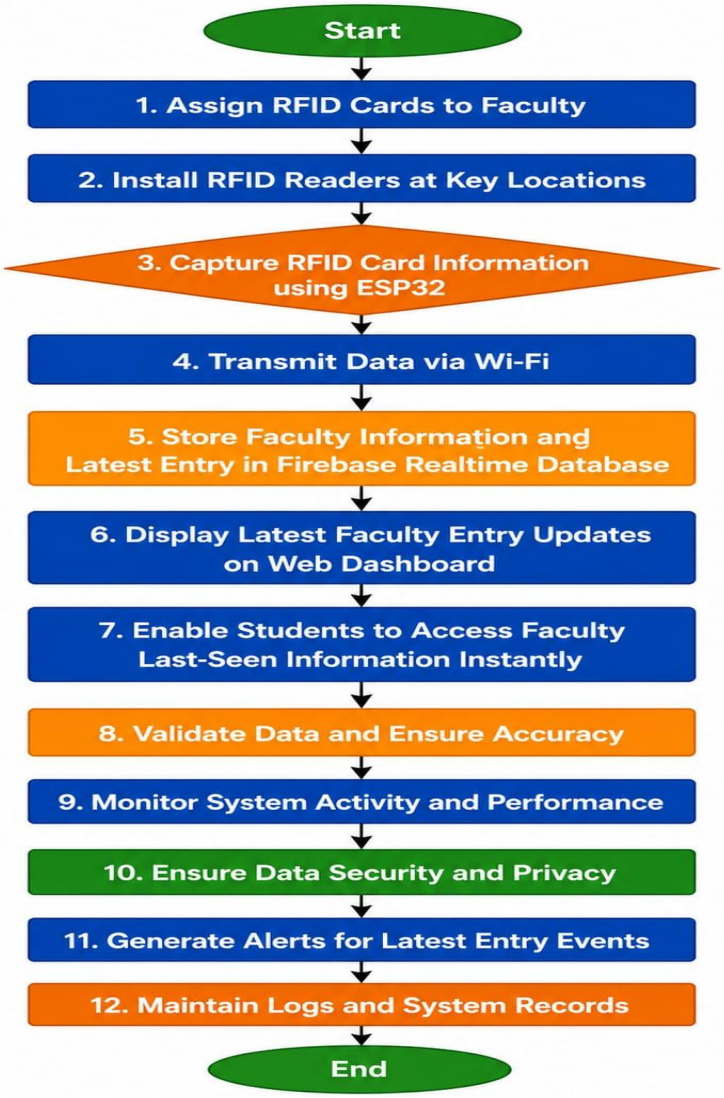
- No more manual searching
- Reduced student waiting time
- Better utilization of time
- Improved academic productivity
- Enhanced user experience

RESULT: With our RFID-Based Faculty Latest Entry Updating System, students can instantly see the faculty member's **last recorded location and last seen time**, reducing time wastage and improving communication and efficiency on campus.

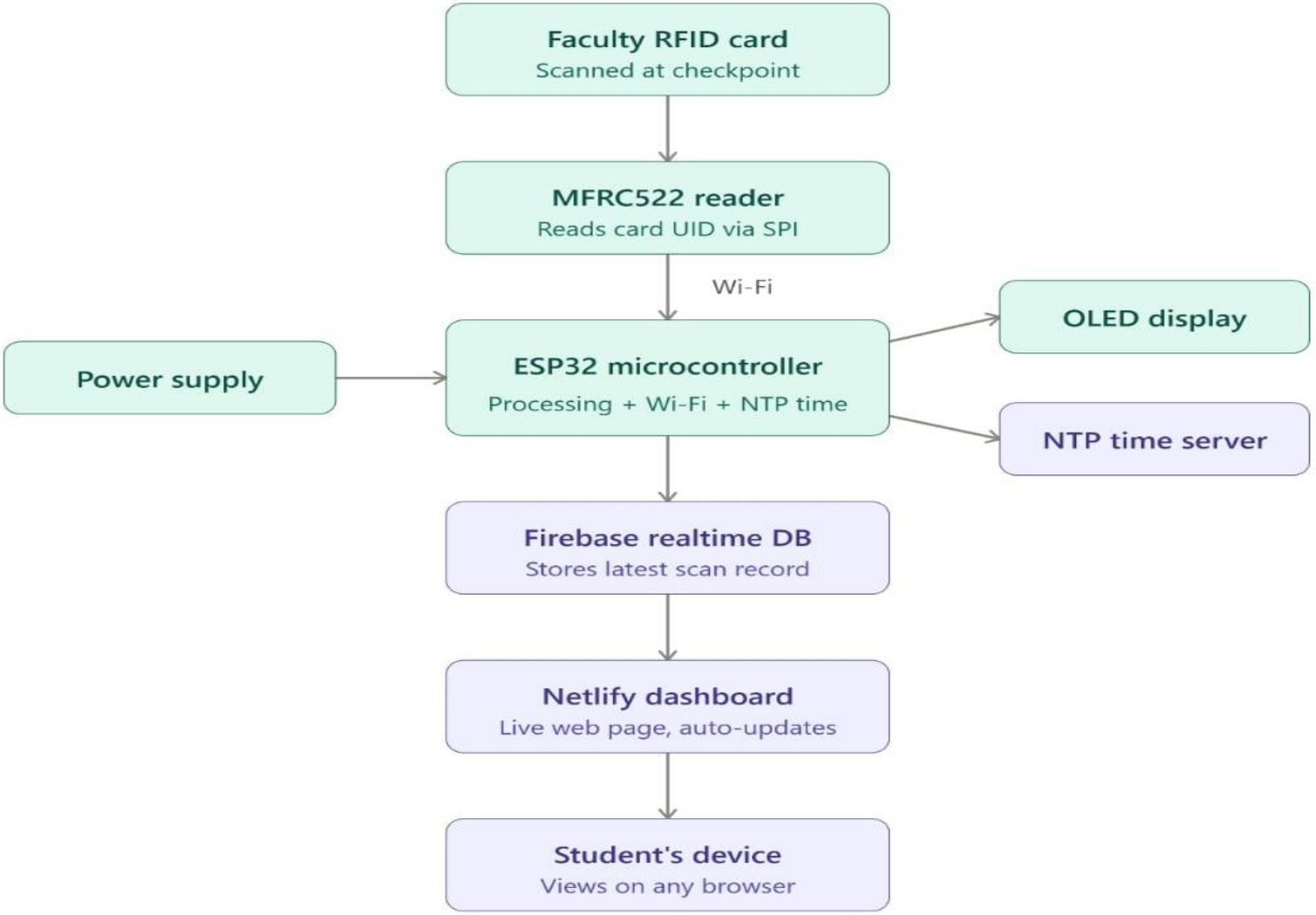
Proposed System



Research Design



BLOCK DAIGRAM



Methodology &Algorithms

Methodology:

- Faculty member scans RFID card at a department.
- RFID reader captures the card UID.
- ESP32 processes the scanned information.
- Current timestamp is obtained from the NTP server.
- Faculty details, latest scanned location, and timestamp are uploaded to Firebase.
- Dashboard updates automatically whenever a new RFID scan is recorded.
- Students can view faculty last-Entry details online.

Algorithm :

- Initialize RFID reader, OLED display, Wi-Fi, and Firebase.
- Wait for RFID card detection.
- Read the RFID card UID.
- Verify UID against registered faculty records.
- Retrieve the current timestamp.
- Update faculty last entry in Firebase.
- Display confirmation message on OLED.
- Refresh dashboard data.
- Return to scanning mode.
- Repeat continuously.

Software / Hardware Descriptions

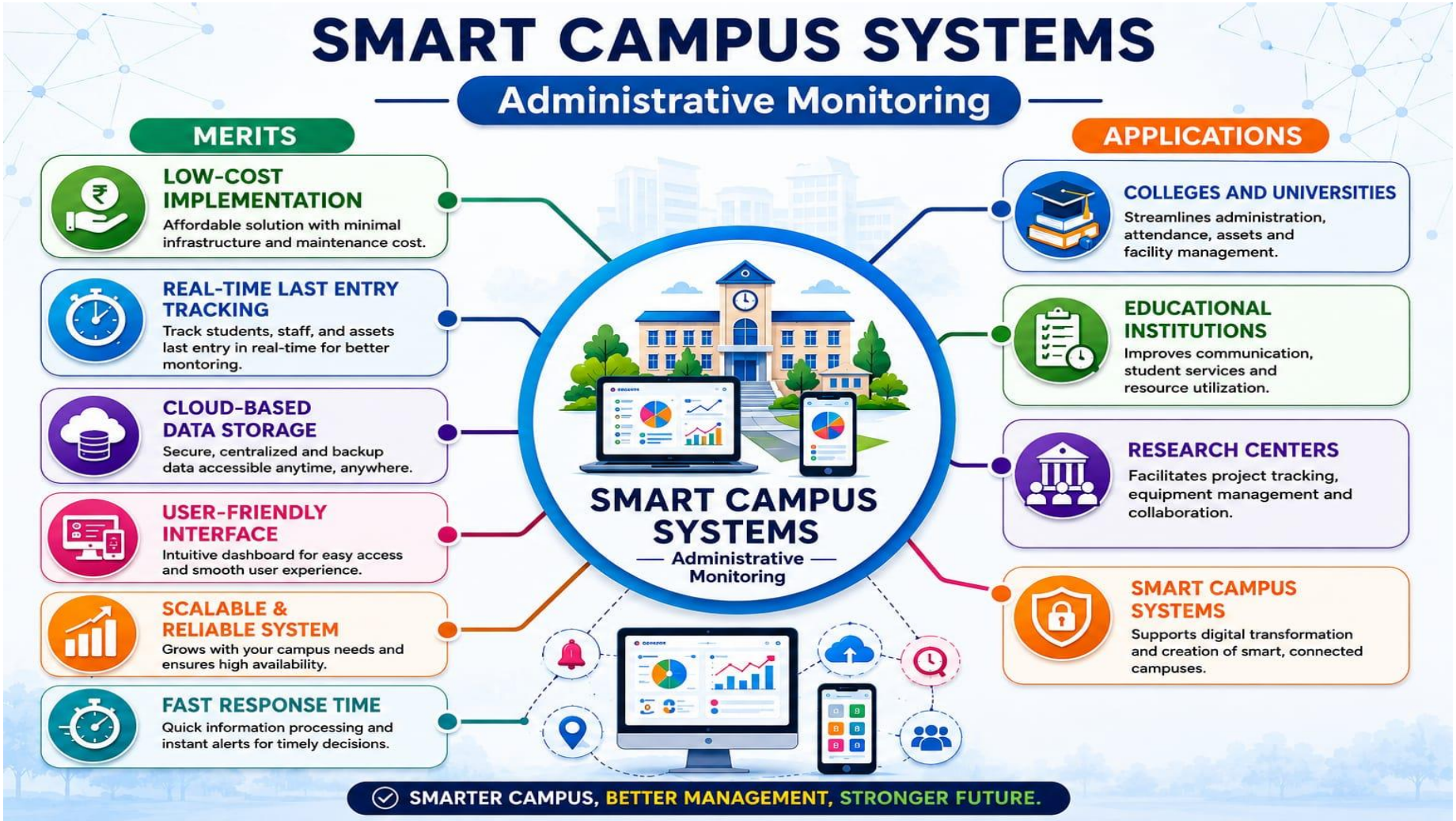
- **Software**

- Visual Studio Code (VS Code)
- PlatformIO IDE Extension
- C++ Programming Language
- Firebase Realtime Database
- HTML Web Dashboard
- Netlify Hosting Platform
- NTP Client

- **Hardware**

ESP32 Development Board (2 Units)
MFRC522 RFID Reader Module (2 Units)
SSD1306 OLED Display
Breadboards
Buck Converters
Charging Modules
Batteries
Brace Wires
Header Pins

Merits / Applications



Experimental Results

EXPERIMENTAL RESULTS

RFID & IoT Based Smart Faculty Tracking System

- RFID cards were **detected and identified** successfully.
- Faculty information was uploaded to **Firebase** in real time.
- Dashboard displayed last **entry of faculty** updates.
- OLED display showed **confirmation messages** after each scan.
- **Latest entry of staff** retrieved instantly.
- Students could view faculty **last seen time instantly.**

SYSTEM WORKFLOW



1. Faculty scans RFID card at a location
2. ESP32 reads card UID and processes data
3. Data is transmitted to Firebase via Wi-Fi
4. Firebase stores faculty details and location
5. Dashboard updates in real time

HARDWARE IN ACTION

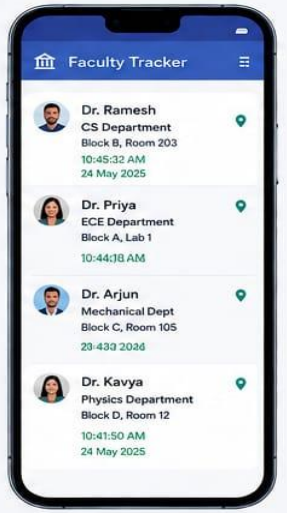


RFID Card Scanning | ESP32 Processing | OLED Confirmation | Scan Confirmed

REAL-TIME DASHBOARD

Faculty Tracking Dashboard

Faculty Name	Department	Location	Last Seen
Dr. Ramesh	Computer Science	CS Department Block B, Room 203	10:45:32 AM 24 May 2025
Dr. Priya	Electronics	ECE Department Block A, Lab 1	10:44:18 AM 24 May 2025
Dr. Arjun	Mechanical	Mechanical Dept Block C, Room 105	10:43:05 AM 24 May 2025
Dr. Kavya	Physics	Physics Department Block D, Room 12	10:41:50 AM 24 May 2025



- **RFID Detection** — Cards detected and identified successfully.
- **Real-Time Upload** — Data uploaded to Firebase in real time.
- **Last Entry Updates** — Dashboard shows last entry of faculty.
- **OLED Messages** — Confirmation messages after each scan.
- **Latest Entry** — Latest entry of staff retrieved instantly.
- **Student Access** — Students view last seen time instantly.

RESULT: The system reliably delivers real-time tracking, accurate last entry updates, and instant access to faculty last seen time, improving transparency, efficiency, and security across the institution.

Conclusion

- RFID-based Faculty Latest Entry Updating System developed successfully.
- Records faculty's last scanned location and timestamp.
- Firebase provides secure cloud-based data management.
- Web dashboard offers easy access to faculty information.
- Reduces student search time and improves communication.
- Cost-effective, scalable, and not a real-time tracking system.

Project Budget

Sl.No.	Parameters	Expenditure
1	Text Books Materials Requirement	-
2	Hardware Materials	2430
3	Industry Visit	-
4	Institutions Visit	-
5	Journal Publication /Conference Presentation	-
6	Printout/Xerox/Binding	100
7	Transportation	-
	Total	2530

References

- ESP32 Technical Documentation
  https://www.geeksforgeeks.org/search/?gq=ESP32+Technical+Documentation
- MFRC522 RFID Reader Datasheet
  https://www.geeksforgeeks.org/search/?gq=RFID%20Reader%20Datasheet
- Firebase Realtime Database Documentation
  https://www.geeksforgeeks.org/search/?gq=Firebase+Realtime+Database+Documentation\
  Visual Studio Code Documentation
  https://www.geeksforgeeks.org/search/?gq=Visual%20Studio%20Code%20Documentation
- Netlify Documentation
  https://www.geeksforgeeks.org/search/?gq=Netlify%20Documentation
- RFID Technology Research PapersIoT-Based Tracking System Journals

Questions ?



Thankyou

